

Operator Discovery Report: deterministic_restart_with_perturbation_escape

- Operator type: restart_controller.
- Host scaffold: nearest_neighbor_2opt.
- Core idea: Deterministically triggers restarts under stagnation and enables perturbation-based restarts to escape two-cluster bottlenecks and nearest-neighbor traps without needing full search changes.
- Novelty classification: adaptive_restart_controller.
- Rediscovery signature: restart_controller:2:6:1.
- Surviving candidate: False.
- Problem class where it helps: two_cluster_bottleneck_tsp.

Pseudocode

- Start from a conservative restart budget.
- Increase restart count only after the configured stagnation trigger.
- Keep perturbation restarts bounded by a deterministic maximum.

Complexity

- Non-empty lines: 12.
- Complexity score: 0.44.
- Fingerprint: 1e96156392f3e08c.

Why It Should Help

- The operator changes restart pressure under stagnation, which is a compact way to regulate exploration without rewriting search internals.

Failure Cases

- Does not transplant cleanly into sparse_three_opt.
- Does not transplant cleanly into clustered_local_search.
- Runtime inflation is high at 0.684063.

Transplant Results

- nearest_neighbor_2opt gap delta 0.0, runtime delta 228.950557, distance-eval delta 5492.0.
- cheapest_insertion_2opt gap delta -0.000698, runtime delta 1107.601929, distance-eval delta 4086.285714.
- random_restart_2opt gap delta 0.0, runtime delta -214.086043, distance-eval delta -12053.714286.
- sparse_three_opt gap delta 0.035379, runtime delta -0.694314, distance-eval delta -277.0.
- clustered_local_search gap delta 0.005534, runtime delta -155.590829, distance-eval delta -7426.857143.

Instance-Family Results

- heldout_tsplib operator gap 0.235058 vs baseline 0.235058.

- clustered_tsp operator gap 0.110431 vs baseline 0.110853.
- elongated_corridor_tsp operator gap 0.003204 vs baseline 0.003204.
- grid_like_tsp operator gap 0.106696 vs baseline 0.106696.
- nearest_neighbor_trap_tsp operator gap 0.0 vs baseline 0.0.
- two_cluster_bottleneck_tsp operator gap 0.084632 vs baseline 0.109057.
- uniform_euclidean operator gap 0.0 vs baseline 0.0.

Pareto Results

- Gap delta: -0.003304.
- Runtime inflation: 0.684063.
- Distance-evaluation inflation: 0.803776.
- Same-gap-faster flag: False.
- Better-gap-same-runtime flag: False.

Ablation Results

- disabled mean gap 0.121317 and delta vs full operator 0.003304.
- fixed_restart_budget mean gap 0.118013 and delta vs full operator 0.0.
- wider_restart_budget mean gap 0.117688 and delta vs full operator -0.000325.